Test Series Code

08

All India Test Series
NEET (2025-26)

SET

01

DURATION: 180 Minutes | DROPPER | **M. MARKS: 720**

DATE: 15/03/2026

SYLLABUS

Physics:	Full Syllabus-11th
Chemistry:	Full Syllabus-11th
Botany:	Full Syllabus-11th
Zoology:	Full Syllabus-11th

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR INSTRUCTIONS:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS): _____

Roll Number: _____

OMR Bar Code Number: _____

Student's Signature: _____ **Invigilator's Signature:** _____

1. A particle moving along a straight line with uniform acceleration has a velocity of 2 m/s at point *A* and 14 m/s at point *B*. If *M* is the midpoint of *AB*, choose the **INCORRECT** statement from the following options.

(1) The velocity at midpoint *M* is 10 m/s.

(2) The average velocity over the distance *AB* is 8 m/s.

(3) The ratio of time taken to cover distance *MB* to the time taken to cover distance *AM* is 1 : 2.

(4) If the distance *AB* is 64 m, the total time taken to cover *AB* is 10 s.

2. Water drops fall from a nozzle at a height *H* above the ground at regular time intervals. When the 1st drop hits the ground, the 4th drop is just leaving the nozzle. At this instant, the height of the 2nd drop from the ground is:

(1) $\frac{5H}{9}$ (2) $\frac{4H}{9}$

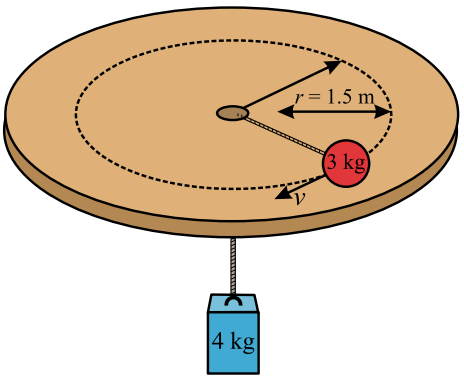
(3) $\frac{7H}{9}$ (4) $\frac{2H}{3}$

3. A circular "Holi Color Sprinkler" ring of mass $M = 5$ kg and radius $R = 2$ m is free to rotate in a horizontal plane about a frictionless axis passing through its center. Four nozzles are attached symmetrically on its outer rim. When the water is turned on, each nozzle ejects colored water tangentially, exerting a constant reaction force of $F = 10$ N opposite to the spray. What is the angular acceleration of the sprinkler ring?

(1) 2 rad/s^2 (2) 4 rad/s^2

(3) 8 rad/s^2 (4) 1 rad/s^2

4. A 3 kg mass is rotating in a horizontal circle of radius 1.5 m on a frictionless table. The mass is connected by a massless inextensible string passing through a frictionless central hole to a larger 4 kg mass hanging below. What linear speed must the 3 kg mass maintain for the 4 kg mass to remain stationary? (Use $g = 10$ m/s^2)



(1) 5 m/s (2) $2\sqrt{5}$ m/s

(3) 4 m/s (4) $\sqrt{15}$ m/s

5. A crate is placed on the flatbed of a truck that is accelerating horizontally on a straight road. The crate does not slip relative to the truck. From the frame of reference of the ground, what is the nature of the work done by the force of static friction exerted by the truck on the crate?

(1) Negative

(2) Positive

(3) Zero

(4) It depends on the mass of the crate.

6. The instantaneous power delivered to a 5 kg particle, starting from rest, is given by the equation $P(t) = 2t$, where P is in watt and t is in second. What is the kinetic energy of the particle at $t = 6$ seconds?

(1) 72 J (2) 36 J

(3) 12 J (4) 18 J

7. A uniform thin bar of length $L = 3$ m is rotated about two different axes perpendicular to the bar. Axis 1 is through the center of the bar, and axis 2 is at a perpendicular distance of $\frac{L}{6}$ from the center. What is the ratio of the moment of inertia about axis 2 to that of axis 1?

(1) $\frac{5}{3}$

(2) $\frac{3}{4}$

(3) $\frac{4}{3}$

(4) $\frac{1}{3}$

[Contd...

8. A juggler maintains 4 balls in motion using one hand, making each of them rise to a height of 20 m from his hand. Each ball is caught and thrown immediately. What is the time interval that he should maintain between the throws of successive balls? (Take $g = 10\ m/s^2$)

(1) 1 s (2) 2 s

(3) 4 s (4) 0.5 s

9. A ship is moving due West with a speed of 10 km/h. A wind is blowing at 10 km/h due North. In which direction will the flag on the mast of the ship flutter?

(1) North-West

(2) North-East

(3) South-East

(4) South-West

10. Two waves are travelling in opposite directions in a taut string. Their equations are $y_1 = 4\sin(2x - 3t)$ and $y_2 = 4\sin(2x + 3t)$, where *x*, *y* are in cm and *t* is in seconds. What is the amplitude of the particle at $x = \frac{\pi}{4}$ cm in the resulting standing wave?

(1) 0 cm (2) 4 cm

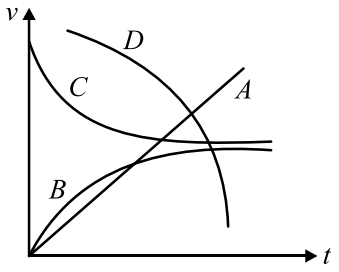
(3) 8 cm (4) $4\sqrt{2}$ cm

11. Two planets P and Q revolve around the Sun in circular orbits. The time period of revolution of planet Q is 8 times that of planet *P*. If v_A represents the areal velocity, what is the ratio of the areal velocity of P to that of Q ($v_{A,P} : v_{A,Q}$)?

(1) 1 : 4 (2) 1 : 2

(3) $1:\sqrt{2}$ (4) 1 : 8

12. A body is thrown vertically downwards in a long column of viscous fluid with a speed greater than the terminal velocity. The best representation of speed of body *v* with time *t* is: (assuming density of body to be greater than that of fluid)



(1) *B* (2) *D*

(3) *C* (4) *A*

13. Two rods *A* and *B* of the same length and cross-sectional area are joined in series. The thermal conductivity of *A* is 2*K* and that of *B* is *K*. The free end of *A* is maintained at 100°C and the free end of *B* at 10°C. In steady state, what is the temperature of the junction?

(1) 70°C

(2) 60°C

(3) 40°C

(4) 55°C

14. A black body at 2000 K emits maximum energy at a wavelength of λ_m. If the temperature is raised to 3000 K, the wavelength corresponding to maximum energy emission will become:

(1) $\frac{3\lambda_m}{2}$ (2) $\frac{2\lambda_m}{3}$

(3) $\frac{4\lambda_m}{9}$ (4) $\frac{9\lambda_m}{4}$

15. A bird of mass 0.5 kg sits on a stretched telegraph wire, causing it to sag. The wire makes an angle of 37° with the horizontal on both sides. If the bird is in equilibrium, the tension in the wire is approximately: ($g = 10\ m\ s^{-2}$)

(1) 2.5 N (2) 5.0 N

(3) 4.2 N (4) 8.3 N

16. Two tuning forks *A* and *B* produce 4 beats per second when sounded together. *A* is in resonance with a 30 cm length of a closed organ pipe, and *B* is in resonance with a 30.5 cm length of the same closed organ pipe (fundamental mode in both). What are the frequencies of tuning fork *A* and *B* respectively?

(1) 244 Hz, 240 Hz

(2) 244 Hz, 248 Hz

(3) 250 Hz, 254 Hz

(4) 255 Hz, 230 Hz

17. An air bubble rises from the bottom of a lake to the surface. If its volume becomes 8 times its initial volume as move from bottom to surface of lake, and the atmospheric pressure is equal to a water column height of *H*, what is the depth of the lake? (Assume temperature is constant).

(1) 7*H* (2) 8*H*

(3) 9*H* (4) *H*

[Contd...

18. In a new system of units, the unit of mass is 10 kg, the unit of length is 1 km, and the unit of time is 1 minute. What is the value of 1 Joule in this new system?

(1) 3.6×10^{-4} units (2) 3.6×10^{-6} units

(3) 2.4×10^{-4} units (4) 6.0×10^{-2} units

19. A particle is projected from the ground with a velocity $\vec{u}=6\hat{i}+8\hat{j}$ m/s. The equation of its trajectory is given by $y=ax-bx^2$, where x and y are in meter. What are the values of a and b? (Take $g=10$ m/s^2).

(1) $a=\frac{4}{3},b=\frac{5}{18}$ (2) $a=\frac{3}{4},b=\frac{5}{36}$

(3) $a=\frac{4}{3},b=\frac{5}{36}$ (4) $a=\frac{3}{4},b=\frac{10}{9}$

20. A particle of mass m is moving in a circular path of constant radius r such that its centripetal acceleration a_c varies with time t as $a_c=k^2rt^2$, where k is a constant. The power delivered to the particle by the forces acting on it is:

(1) mk^2r^2t (2) mkr^2t^2

(3) zero (4) $mk^2r^2t^2$

21. From a uniform circular disc of radius R, a circular hole of radius $\frac{R}{2}$ is cut out. The center of the hole is at a distance $\frac{R}{2}$ from the center of the original disc. The center of mass of the remaining portion shifts by:

(1) $\frac{R}{6}$ (2) $\frac{R}{3}$

(3) $\frac{R}{2}$ (4) $\frac{R}{4}$

22. A particle is fired vertically upward with a speed of $v=\sqrt{gR}$ from the surface of Earth (Radius R). The maximum height attained by the particle above the surface is:

(1) R (2) $\frac{R}{2}$

(3) $2R$ (4) $\frac{R}{3}$

23. A steel wire of length L and cross-section A is stretched by a force F, extending by l. If another wire of the same material but length $2L$ and cross-section $2A$ is stretched by force $2F$, the extension will be:

(1) l (2) $2l$

(3) $\frac{l}{2}$ (4) $4l$

24. A particle executes simple harmonic motion with an amplitude of 4 cm. At the mean position, the velocity is 10 cm/s. What is the distance of the particle from the mean position when its speed becomes 5 cm/s?

(1) $2\sqrt{3}$ cm (2) 2 cm

(3) $\sqrt{3}$ cm (4) 3 cm

25. A hypothetical linear temperature scale, denoted by °P, is defined such that the freezing point of water is $-10°P$ and the boiling point of water is $140°P$. A certain metal has a specific heat capacity of 900 J/(kg·K) in the SI system. What is the specific heat capacity of this metal expressed in J/(kg·°P)?

(1) 1350
(2) 600
(3) 540
(4) 1200

26. Rain is falling vertically with a speed of 30 m/s. A woman rides a bicycle with a speed of 10 m/s in the North to South direction. What is the direction in which she should hold her umbrella to protect herself from the rain?

(1) $\tan^{-1}(3)$ with vertical

(2) $\tan^{-1}\left(\frac{1}{3}\right)$ with vertical

(3) $\tan^{-1}\left(\frac{2}{3}\right)$ with horizontal

(4) $\tan^{-1}\left(\frac{1}{4}\right)$ with horizontal

27. The root mean square speed of hydrogen molecules at a certain temperature is v. If the temperature is doubled and hydrogen gas dissociates into atomic hydrogen, the rms speed will become:

(1) v (2) $2v$

(3) $\sqrt{2}v$ (4) $4v$

[Contd...

28. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:

Assertion (A): It is easier to pull a lawn roller than to push it.

Reason (R): Pushing increases the apparent weight and hence the frictional force, while pulling decreases the apparent weight and frictional force.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true and R is NOT the correct explanation of A.

29. A string of length 1 m and mass 5 g is fixed at both ends. The tension in the string is 8.0 N. The fundamental frequency of vibration is:

(1) 100 Hz (2) 20 Hz

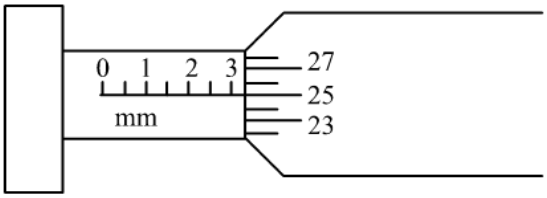
(3) 40 Hz (4) 50 Hz

30. A particle of mass $2m$ moving with speed u collides elastically with another stationary particle of mass $2m$. If the collision is oblique, the angle between the velocities of the two particles after collision is:

(1) 45° (2) 90°

(3) 60° (4) 180°

31. The least count of screw gauge shown in figure is 0.01 mm. The value of reading shown on instrument is: (assume no zero error in instrument)



(1) 3.25 mm

(2) 3.24 mm

(3) 3.30 mm

(4) 3.26 mm

32. A Carnot engine works between temperatures 727°C and 27°C. The efficiency of the engine is:

(1) 30 % (2) 70 %

(3) 96 % (4) 10 %

33. A stone of mass 1 kg tied to a light string of length $L = \frac{10}{3}$ m is whirling in a vertical circle. If the ratio of maximum tension to minimum tension in the string is 4, what is the speed of the stone at the highest point? ($g = 10$ m/s^2).

(1) 10 m/s (2) $5\sqrt{2}$ m/s

(3) $10\sqrt{3}$ m/s (4) 20 m/s

34. A clock with a metal pendulum keeps correct time at 20°C. If the temperature rises to 40°C, the clock will: (Coefficient of linear expansion $\alpha = 10^{-5}$ / °C)

(1) Lose 8.64 s per day

(2) Gain 8.64 s per day

(3) Lose 4.32 s per day

(4) Gain 4.32 s per day

35. The fractional decrease in the volume of oil is 1 percent when a pressure of 2×10^7 N/m^2 is applied. The bulk modulus of the oil and its compressibility are respectively:

(1) 2×10^9 N/m^2, 5×10^{-10} m^2 / N

(2) 2×10^9 N/m^2, 5×10^{-9} m^2 / N

(3) 10^9 N/m^2, 2×10^{-10} m^2 / N

(4) 5×10^9 N/m^2, 2×10^{-10} m^2 / N

36. In a laboratory experiment to determine the resistance $R = \frac{V}{I}$, the voltage changes from V to $V + \Delta V$ and the current changes from I to $I - \Delta I$. Given that the fractional changes are $\frac{\Delta V}{V} = 0.5$ and $\frac{\Delta I}{I} = 0.25$, what is the percentage change in the resistance?

(1) 75% (2) 100%

(3) 125% (4) 50%

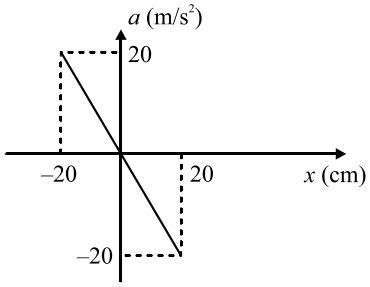
37. The acceleration due to gravity at a distance $r = \frac{R}{6}$ from the center of the Earth (where R is Earth's radius) is g'. If the acceleration due to gravity on the surface is g, then the value of g' is:

(1) $\frac{g}{6}$ (2) $\frac{5g}{6}$

(3) $\frac{g}{36}$ (4) $6g$

[Contd...

38. A body of mass 2 kg executes SHM of amplitude 20 cm about origin. The acceleration a versus displacement x graph is as shown below. Angular frequency of the body is:



(1) 40 rad s^{-1} (2) 20 rad s^{-1}
(3) 10 rad s^{-1} (4) 5 rad s^{-1}

39. Given below are two statements:

Statement I: The work done by a conservative force is independent of the path taken between the initial and final position.

Statement II: Kinetic energy of body is independent of the reference frame.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement I and Statement II both are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Statement I and Statement II both are incorrect.

40. A U-tube manometer contains mercury of density 13.6 g/cc. The left arm is connected to a gas container and the right arm is open to the atmosphere. The mercury level in the right arm is 10 cm higher than the level in the left arm. If the atmospheric pressure is 76 cm of Hg, what is the absolute pressure of the gas?

(1) 66 cm of Hg
(2) 86 cm of Hg
(3) 76 cm of Hg
(4) 10 cm of Hg

41. A 3 m long uniform ladder of mass 20 kg leans on a frictionless wall. Its feet rests on the rough floor 1 m from the wall. The normal forces of the wall and the floor respectively will be nearly: (g = 10 m s^{-2})

(1) 35 N and 200 N
(2) 30 N and 180 N
(3) 25 N and 200 N
(4) 45 N and 160 N

42. One mole of an ideal monoatomic gas undergoes a thermodynamic process in which the pressure P and volume V are related by the equation $P = kV^2$, where k is a positive constant. If the temperature of the gas is increased by ΔT, the amount of heat supplied to the gas is:

(1) $\frac{11}{6}R\Delta T$ (2) $\frac{5}{2}R\Delta T$
(3) $\frac{4}{3}R\Delta T$ (4) $2R\Delta T$

43. A person of mass m is standing on a weighing machine inside a lift. Match the state of motion of the lift in **List-I** with the corresponding Normal Reaction (N) exerted by the machine in **List-II**.

List-I (Motion of Lift)		**List-II (Normal Reaction N)**	
(A)	Accelerating upwards with a	(I)	$N = mg$
(B)	Accelerating downwards with 'a' ($a < g$)	(II)	$N = m(g + a)$
(C)	Moving with constant velocity	(III)	$N = 0$
(D)	Free fall (cable breaks)	(IV)	$N = m(g - a)$

Choose the **correct** answer from the options given below:

(1) A-II, B-IV, C-I, D-III
(2) A-I, B-IV, C-II, D-III
(3) A-I, B-IV, C-III, D-II
(4) A-IV, B-I, C-III, D-II

44. The property of matter by virtue of which a body tends to regain its original shape and size after removal of deforming force is called:

(1) plasticity (2) elasticity
(3) ductility (4) compressibility

45. A planet in a distant solar system is 8 times massive than the earth and its radius is 8 times smaller. Given that the escape velocity from the earth surface is 11 km s^{-1}, the escape velocity from the surface of the planet would be:

(1) 8.8 km s^{-1} (2) 9.8 km s^{-1}
(3) 88 km s^{-1} (4) 880 km s^{-1}

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46. In the seventh period, the orbitals that are filled are: (Aufbau sequence)
(1) 7s, 6f, 5d, 7p
(2) 7s, 5f, 6d, 7p
(3) 7s, 6d, 5f, 7p
(4) 7s, 5d, 6f, 7p

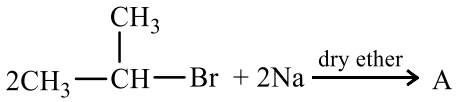
47. The atomic number of the element from the following with lowest 1st ionisation enthalpy is:
(1) 12 (2) 38
(3) 55 (4) 17

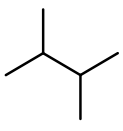
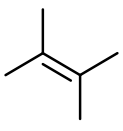
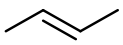
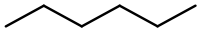
48. In the following, the correct bond order sequence is:
(1) $NO > NO^+ > NO^-$
(2) $NO^+ > NO > NO^-$
(3) $NO^- > NO > NO^+$
(4) $NO^- > NO^+ > NO$

49. Which of the following are isostructural pairs?
(A) NO_2^- and SO_2
(B) NH_4^+ and BF_4^-
(C) CO_2 and SO_2
(D) PCl_5 and SF_4
(1) C and D only
(2) A and B only
(3) A and C only
(4) B and C only

50. Choose the **correct** option for free expansion of an ideal gas into vacuum in a thermally insulated vessel:
(1) $q = 0, w \neq 0, \Delta T \neq 0$
(2) $q \neq 0, w = 0, \Delta T = 0$
(3) $q = 0, w = 0, \Delta U = 0, \Delta T = 0$
(4) $q = 0, w = 0, \Delta U \neq 0, \Delta T \neq 0$

51. The major product obtained in the reaction given below is:



(1) 
(2) 
(3) 
(4) 

52. Which of the following compound **cannot** act as a Lewis base?
(1) BF_3 (2) PH_3
(3) H_2S (4) ClF_3

53. The **correct** orders among the following for Be, Mg, Ca, Sr, Ba are:
(A) Atomic radius: Be < Mg < Ca < Sr < Ba
(B) 1st Ionisation Energy: Be > Mg > Ca > Sr > Ba
(C) Basicity of hydroxides: $Ba(OH)_2 < Sr(OH)_2 < Ca(OH)_2 < Mg(OH)_2 < Be(OH)_2$
(D) Thermal stability of carbonates: $BeCO_3 > MgCO_3 > CaCO_3 > SrCO_3 > BaCO_3$
Choose the **correct** answer from the options given below:
(1) B and D Only (2) A and C Only
(3) C and D Only (4) A and B Only

54. Arrange the following bonds according to their average bond energies in descending order:
H–Cl, H–Br, H–F, H–I
(1) H–F > H–Cl > H–Br > H–I
(2) H–I > H–Br > H–Cl > H–F
(3) H–Cl > H–Br > H–I > H–F
(4) H–Br > H–I > H–Cl > H–F

55. Geometrical isomerism is shown by:
(1) But-2-ene
(2) 1,2-Dichloroethene
(3) 3,4-Dimethylhex-3-ene
(4) All of these

56. Within each pair of elements Cl & Br, O & S and Na & K, respectively, the elements that release more energy upon an electron gain are:
(1) Br, O and K (2) Cl, S and Na
(3) Cl, O and K (4) Br, S and Na

57. Match **List-I** with **List-II**.

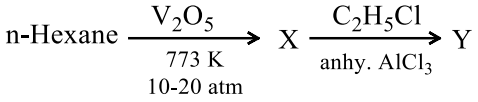
List-I (Spectral Series for Hydrogen)		**List-II (Spectral Region/ Higher Energy State)**	
(A)	Lyman	(I)	Infrared region
(B)	Balmer	(II)	UV region
(C)	Paschen	(III)	Visible region

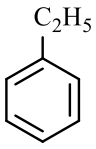
Choose the **correct** answer from the options given below:
(1) A-II, B-I, C-III
(2) A-I, B-III, C-II
(3) A-II, B-III, C-I
(4) A-I, B-II, C-III

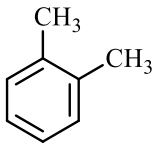
58. The **correct** order of acidity is:
(1) $H_2C{=}CH_2 > CH_3{-}CH_3 > CH{\equiv}CH$
(2) $CH{\equiv}CH > H_2C{=}CH_2 > CH_3{-}CH_3$
(3) $CH{\equiv}CH > CH_3{-}CH_3 > CH_2{=}CH_2$
(4) $CH_3{-}CH_3 > CH_2{=}CH_2 > CH{\equiv}CH$

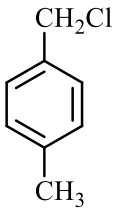
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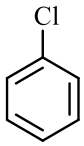
59. Identify Y in the given reaction:



(1) 

(2) 

(3) 

(4) 

60. Given below are two statements:

Statement-I: In Bohr's model, the electron's angular momentum in a stationary orbit satisfies $mvr = n\hbar$.

Statement-II: A fixed circular orbit assigns definite r and p simultaneously, conflicting with Heisenberg uncertainty.

Choose the *most appropriate* answer:

(1) Both Statement-I and Statement-II are correct
(2) Statement-I is correct but Statement-II is incorrect
(3) Statement-I is incorrect but Statement-II is correct
(4) Both Statement-I and Statement-II are incorrect

61. 200 g aqueous solution of D-glucose contains 12.0% carbon by mass. The molality of the solution is:

(1) 1.19 (2) 2.38
(3) 3.57 (4) 4.76

62. Identify the **correct** statements about p-block elements and their compounds.

(A) Nonmetals generally have higher electronegativity than metals.
(B) Nonmetals generally have lower ionisation enthalpy than metals.
(C) Compounds between highly reactive metals and highly reactive nonmetals are generally ionic.
(D) Nonmetal oxides are generally basic in nature.
(E) Metal oxides are generally acidic or neutral in nature.

(1) (D) and (E) only
(2) (A) and (C) only
(3) (B) and (E) only
(4) (B) and (D) only

63. The ionic radius of Na^+ is 1.00 Å. The ionic radii (in Å) of Mg^{2+} and Al^{3+}, respectively, are:

(1) 0.85 and 0.90
(2) 0.72 and 0.54
(3) 1.05 and 0.72
(4) 0.68 and 0.90

64. Number of molecules/ions from the following in which the central atom is involved in sp^3 hybridization is CO_3^{2-}, BF_3, BrO_2^-, BrO_3^-

(1) 1 (2) 2
(3) 3 (4) 4

65. The standard reaction Gibbs energy at temperature T is: $\Delta_r G^\circ = A - BT$, where A and B are nonzero constants. Which statement is **true**?

(1) Endothermic if $A > 0$
(2) Exothermic if $A > 0$ and $B < 0$
(3) Endothermic if $A < 0$ and $B > 0$
(4) Exothermic if $B < 0$

66. For the reaction, $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, $\Delta H = -92\ kJ\ mol^{-1}$ and $K_c = 1.0 \times 10^5$.

Which statement is **incorrect**?

(1) Large K_c implies reaction goes to completion, so no catalyst is required.
(2) Increasing pressure shifts equilibrium in the forward direction.
(3) K_c decreases as temperature increases.
(4) Adding inert gas at constant volume does not change K_c.

67. In which one of the following sets do all species show disproportionation reaction?

(1) ClO^-, H_2O_2, I_2, Cu^+
(2) ClO^-, F_2, H_2O_2, Cu^+
(3) IO_3^-, H_2O_2, I_2, Cu^+
(4) Cl_2, IO_3^-, Cu^{2+}, H_2O_2

68. Group-13 elements react with O_2 in amorphous form to form oxides of type M_2O_3. Which among the following is the most basic oxide?

(1) B_2O_3
(2) Al_2O_3
(3) In_2O_3
(4) Tl_2O_3

[Contd...

69. Given below are two statements:
Statement-I: Hyperconjugation is not a permanent effect.
Statement-II: Generally, more alkyl groups on a positively charged carbon increase hyperconjugation and stabilise the cation.
Choose the **correct** answer:
(1) Statement-I is correct but Statement-II is incorrect
(2) Both Statement-I and Statement-II are incorrect
(3) Statement-I is incorrect but Statement-II is correct
(4) Both Statement-I and Statement-II are correct

70. The adsorbent used in adsorption chromatography is/are
(A) silica gel (B) alumina
(C) quick lime (D) magnesia
Choose the *most appropriate* answer:
(1) B only (2) C and D only
(3) A and B only (4) D only

71. Kjeldahl's method **cannot** be used to estimate nitrogen for which compound?
(1) CH_3CONH_2 (2) $C_6H_5CH_2NH_2$
(3) $C_6H_5NO_2$ (4) $CH_3CH_2NH_2$

72. The set of meta-directing functional groups is:
(1) $-CN$, $-NH_2$, $-OCH_3$, $-R$
(2) $-NO_2$, $-NH_2$, $-COOH$, $-COOR$
(3) $-NO_2$, $-CHO$, $-SO_3H$, $-COR$
(4) $-CN$, $-CHO$, $-NHCOCH_3$, $-CH_3$

73. The alkane from below having two secondary hydrogens is:
(1) 2,2,4-Trimethylpentane
(2) 2,3-Dimethylbutane
(3) 3-Methylhexane
(4) 2,2,3-Trimethylbutane

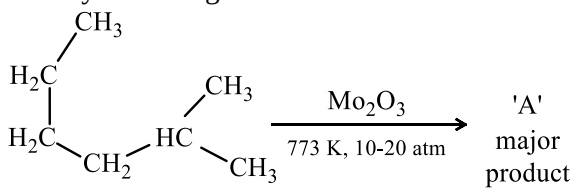
74. Evaluate the following statements for group 14 elements:
(A) Covalent radius decreases down the group from C to Pb regularly.
(B) Electronegativity decreases gradually from C to Pb down the group.
(C) Maximum covalence of C is 4 whereas others can expand covalence due to d orbitals.
(D) Heavier elements do not form $p\pi - p\pi$ bonds.
(E) Carbon can exhibit negative oxidation states.
Choose the **correct** answer:
(1) (C), (D) and (E) only
(2) (A) and (B) only
(3) (A), (B) and (C) only
(4) (C) and (D) only

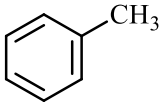
75. During which of the following processes does entropy decrease?
(A) Freezing of water to ice at $0^\circ C$
(B) Freezing of water to ice at $-10^\circ C$
(C) $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$
(D) Adsorption of $CO(g)$ on a metal surface
(E) Dissolution of NaCl in water
Choose the **correct** answer:
(1) (A) and (E) only
(2) (A), (B), (C) and (D) only
(3) (B), (C) and (E) only
(4) (A), (C) and (D) only

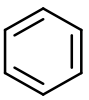
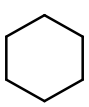
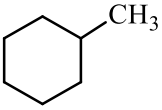
76. $IO_3^- + xI^- + 6H^+ \rightarrow 3I_2 + 3H_2O$. What is the value of x?
(1) 3
(2) 4
(3) 5
(4) 6

77. For a concentrated solution of weak electrolyte A_2B_3 of concentration c, the degree of dissociation α is:
(1) $\left(\frac{K_{eq}}{108c^4}\right)^{1/5}$
(2) $\left(\frac{K_{eq}}{54c^4}\right)^{1/5}$
(3) $\left(\frac{K_{eq}}{108c^5}\right)^{1/5}$
(4) $\left(\frac{K_{eq}}{27c^4}\right)^{1/5}$

78. Identify A in the given chemical reaction.



(1) 

(2) 
(3) 
(4) 

[Contd...

79. The **correct** statement/s about hydrogen bonding is/are:
(A) Hydrogen bonding is possible when H is covalently bonded to N, O or F.
(B) Intermolecular H-bonding is present in o-nitrophenol.
(C) Intramolecular H-bonding is present in HF.
(D) The strength of H-bonding depends on physical state/association.
(E) H-bonding strongly affects structure and properties.
Choose the **correct** answer:
(1) A only (2) A, D, E only
(3) A, B, D only (4) A, B, C only

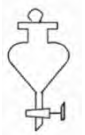
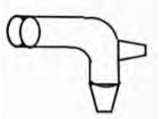
80. Inert gases have positive electron gain enthalpy. Its correct order is:
(1) $He < Xe < Ar \approx Kr < Ne$
(2) $He < Ne < Ar < Kr < Xe$
(3) $Ne < He < Ar < Kr < Xe$
(4) $Ar < Kr < Xe < Ne < He$

81. The **correct** statement about probability density is: it becomes zero at some finite non-zero distance from nucleus for:
(1) 1s orbital (2) 2p orbital
(3) 2s orbital (4) 3d orbital

82. If 5 moles of $BaCl_2$ is mixed with 3 moles of Na_3PO_4, the maximum number of moles of $Ba_3(PO_4)_2$ formed is:
(1) 1 (2) 1.5
(3) 3 (4) 4

83. If the average molar mass of chlorine is $35.2\ g/mol$. The ratio of ^{35}Cl to ^{37}Cl in natural chlorine is close to:
(1) 9:1 (2) 7:3
(3) 3:1 (4) 1:1

84. Lead exists mainly in +2 and +4 oxidation states, with +2 being more stable. This is due to:
(1) Inert pair effect (2) Diagonal relationship
(3) Lattice effect (4) Lanthanoid contraction

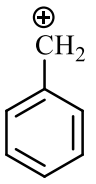
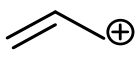
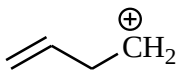
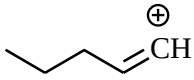
85. Which of the following is structure of a separating funnel?
(1)  (2) 
(3)  (4) 

86. The redox reaction among the following is:
(1) Reaction of $[Co(NH_3)_6]Cl_3$ with $AgNO_3$
(2) Formation of ozone from atmospheric oxygen in sunlight
(3) Combination of dinitrogen with dioxygen at high temperature to form NO
(4) Reaction of H_2SO_4 with NaOH

87. An acidic buffer is obtained on mixing:
(1) $100\ mL\ 0.1\ M\ HCl + 200\ mL\ 0.1\ M\ NaCl$
(2) $200\ mL\ 0.2\ M\ HCl + 200\ mL\ 0.2\ M\ CH_3COONa$
(3) $100\ mL\ 0.2\ M\ CH_3COOH + 50\ mL\ 0.2\ M\ NaOH$
(4) $100\ mL\ 0.2\ M\ CH_3COOH + 200\ mL\ 0.2\ M\ NaOH$

88. The ratio $\frac{K_p}{K_c}$ for the reaction $N_2O_4(g) \rightleftharpoons 2NO_2(g)$ is:
(1) $\frac{1}{\sqrt{RT}}$ (2) RT
(3) 1 (4) $\frac{1}{RT}$

89. 24.0 L of CO_2 is produced on complete combustion of 12.0 L gaseous mixture of ethene and methane at $25^\circ C$, $1\ atm$. Heat evolved is ———— kJ.
Given: $\Delta H_c(CH_4) = -900\ kJ\ mol^{-1}$,
$\Delta H_c(C_2H_4) = -1400\ kJ\ mol^{-1}$, molar volume $= 24\ L\ mol^{-1}$.
(1) 450 (2) 700
(3) 900 (4) 1400

90. Among the given species the Resonance stabilized carbocations are:
(A) 
(B) 
(C) 
(D) 
(1) (A), (B) and (C) only
(2) (C) and (D) only
(3) (A), (B) and (D) only
(4) (A) and (B) only

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91. Identify the **incorrect** statement about 'reductionist biology' approach to study and understand living organisms.
(1) It cannot explain the abnormal processes that occur during any diseased condition.
(2) It uses concepts and techniques of physics and chemistry to understand biology.
(3) It is a physicochemical approach.
(4) It uses cell-free systems for investigation.

92. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Zygotene	(I)	Terminalisation of chiasmata
(B)	Pachytene	(II)	Synapsis
(C)	Diplotene	(III)	Crossing over
(D)	Diakinesis	(IV)	Formation of Chiasmata

Choose the *most appropriate* answer from the options given below.
(1) A-I, B-III, C-II, D-IV
(2) A-II, B-IV, C-III, D-I
(3) A-III, B-II, C-IV, D-I
(4) A-II, B-III, C-IV, D-I

93. In monocotyledonous plants, the primary root is;
(1) retained throughout the life of the plant.
(2) formed by the direct elongation of plumule.
(3) short-lived.
(4) replaced by aerial roots.

94. The primary CO_2 acceptor in Hatch and Slack pathway;
(1) is a 3-C molecule.
(2) is RuBP.
(3) is a 5-C sugar.
(4) oxidises it.

95. Family Convolvulaceae belongs to the order;
(1) Polymoniales. (2) Poales.
(3) Sapindales. (4) Primata.

96. If fatty acids were to be respired, they would first be degraded to which compound before entering the respiratory pathway?
(1) PGAL (2) Pyruvate
(3) Acetyl CoA (4) Citric acid

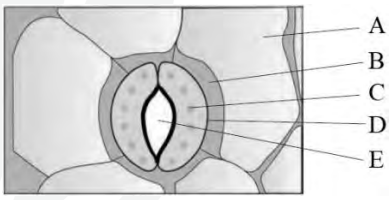
97. Identify the correct statements regarding Viruses.
A. The name virus that means venom or poisonous fluid was given by Dmitri Ivanowsky.
B. Viruses are facultative parasites.
C. W.M. Stanley showed that viruses could be crystallised and crystals consist largely of proteins.
D. In general, viruses that infect plants have single stranded RNA.
E. Bacterial viruses or bacteriophages are usually single stranded DNA viruses.
Choose the **correct** answer from the options given below:
(1) A, C and E only (2) C and D only
(3) A, B and C only (4) B, D and E only

98. Match **List-I** with **List-II**.

List-I (Alga)		List-II (Associated characteristic)	
(A)	*Volvox*	(I)	Form massive plant bodies
(B)	*Ectocarpus*	(II)	Colonial form
(C)	Kelps	(III)	Filamentous form
(D)	*Porphyra*	(IV)	Store food in the form of Floridean starch

Choose the *most appropriate* answer from the options given below.
(1) A-I, B-III, C-II, D-IV
(2) A-II, B-I, C-III, D-IV
(3) A-III, B-II, C-IV, D-I
(4) A-II, B-III, C-I, D-IV

99. Choose the **correct** statement for the diagram shown below.



(1) It represents the stomatal apparatus in an isobilateral leaf.
(2) B, D and E are the part of stomatal apparatus.
(3) 'A' are the endodermal cells.
(4) The organelle 'C' is a part of endomembrane system.

100. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): Removal of root tips (decapitation) usually results in the growth of lateral buds.
Reason (R): The growing apical bud inhibits the growth of the lateral (axillary) buds.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

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101. Given below are two statements:

Statement I: The outer nuclear membrane is usually continuous with the endoplasmic reticulum and also bears ribosomes on it.

Statement II: Microvilli are present in animal and plant cells.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

102. Who among the following was awarded with the 'triple crown of biology'?

(1) Ernst Mayr
(2) Carolus Linnaeus
(3) Katherine Esau
(4) Melvin Calvin

103. The reaction centre in PS II is one molecule of;

(1) chlorophyll *a*. (2) chlorophyll *b*.
(3) carotenoid. (4) xanthophyll.

104. Which of the following statements regarding Glycolysis are **incorrect**?

A. It is the only process in respiration in anaerobic organisms.
B. Glucose is completely oxidized during this process.
C. ATP is utilised in the conversion of Fructose 6-phosphate to Fructose 1, 6-bisphosphate.
D. One molecule of NADH + H^+ is formed when 3-phosphoglyceric acid (PGA) is converted to 1, 3-bisphosphoglycerate (BPGA).
E. The pyruvic acid is the key product of glycolysis.

Choose the **correct** answer from the options given below:

(1) B and C
(2) B and D
(3) C and E
(4) A and E

105. In which of the following plants, the margins of sepals or petals overlap one another but not in any particular direction?

(1) *Calotropis* (2) Cotton
(3) *Cassia* (4) Pea

106. Select the **incorrectly** matched pair.

(1) Biological names - Generally in Latin and written in italics
(2) First word in biological name - Represents the specific epithet
(3) Second component in biological name - Starts with a small letter
(4) Handwritten biological names - Separately underlined

107. The cell walls of _____ form two thin overlapping shells, which fit together as in a soap box.

(1) diatoms (2) dinoflagellates
(3) bacteria (4) plants

108. Read the following statements about Xylem:

A. Gymnosperms lack vessels in their xylem.
B. Tracheids are elongated or tube like cells with thick and lignified walls and tapering ends.
C. Vessel members are interconnected through perforations in their common walls.
D. Xylem parenchyma cells are dead and thin-walled.
E. In roots, the protoxylem lies towards the centre (pith) and the metaxylem lies towards the periphery.

The **correct** statements are;

(1) A, B and C (2) B, C and D
(3) A, C and E (4) A, D and E

109. A cell (16 chromosomes and 20 pg of DNA in G_1 phase) undergoes division and produces daughter cells with 8 chromosomes each. The expected amount of DNA in each daughter cell would be;

(1) 40 pg (2) 20 pg
(3) 30 pg (4) 10 pg

110. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): The spread of living pteridophytes is limited and restricted to narrow geographical regions.

Reason (R): The gametophytes of pteridophytes require cool, damp, shady places to grow and need water for fertilisation.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

111. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Respiratory climactic	(I)	Auxins
(B)	Bolting	(II)	Gibberellins
(C)	Stress hormone	(III)	Ethylene
(D)	Apical dominance	(IV)	Abscisic acid

Choose the *most appropriate* answer from the options given below.
(1) A-II, B-III, C-IV, D-I
(2) A-III, B-II, C-IV, D-I
(3) A-III, B-I, C-II, D-IV
(4) A-IV, B-II, C-I, D-III

112. During non-cyclic photophosphorylation;
(1) $NADPH + H^+$ is not synthesised.
(2) the pH of thylakoid lumen increases.
(3) both PS I and PS II are involved.
(4) the electrons lost from PS I are compensated by electrons from water splitting.

113. In the mitochondrial ETS, the catalytic site of ATP synthase enzyme;
(1) is contained in the F_0 component.
(2) is present within the integral membrane protein complex.
(3) converts ATP into ADP.
(4) is present towards the matrix.

114. Which of the following statements regarding prokaryotic cells are **correct**?
A. The genetic material is basically naked, not enveloped by a nuclear membrane.
B. The genomic DNA is always linear.
C. Many bacteria have small circular DNA outside the genomic DNA called plasmids.
D. A specialised differentiated form of cell membrane called mesosome is characteristic of prokaryotes.
E. Ribosomes are absent in prokaryotic cells.
Choose the **correct** answer from the options given below:
(1) A, C and D only
(2) A, B and C only
(3) B, D and E only
(4) C, D and E only

115. Given below are two statements:
Statement I: A flower is a modified shoot wherein the shoot apical meristem changes to floral meristem.
Statement II: When a shoot tip transforms into a flower, it is always solitary.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

116. The bread mould;
(1) is a unicellular fungus.
(2) cannot reproduce sexually.
(3) is classified in the class phycomycetes.
(4) is *Albugo*.

117. Select the **correctly** matched pair.
(1) Parenchyma - Thick walled and dead cells
(2) Collenchyma - Thickened at corners due to cellulose, hemicellulose and pectin
(3) Sclerenchyma - Living cells with protoplasts
(4) Xylem fibres - Obliterated central lumens and always septate

118. Arrange the following taxonomic categories in descending order:
A. Family B. Class
C. Species D. Kingdom
E. Genus
Choose the **correct** answer from the options given below:
(1) C → E → A → B → D
(2) A → B → C → D → E
(3) B → D → E → C → A
(4) D → B → A → E → C

119. Given below are two statements:
Statement I: Mitosis usually results in the production of diploid daughter cells with identical genetic complement.
Statement II: Cell growth results in disturbing the ratio between the nucleus and the cytoplasm, thus cell divides to restore the nucleo-cytoplasmic ratio.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

120. Identify the **incorrect** statement regarding gymnosperms.
(1) The ovules are not enclosed by any ovary wall and remain exposed.
(2) The roots are generally tap roots.
(3) The male and female gametophytes have an independent free-living existence.
(4) The stems are unbranched in *Cycas* and branched in *Pinus*.

121. A farmer wants to induce flowering in pineapples and synchronise fruit-set. Which plant growth regulator should be applied?
(1) Auxin (2) Gibberellin
(3) Cytokinin (4) Ethylene

122. The major limiting factor for photosynthesis is;
(1) water. (2) CO_2.
(3) temperature. (4) light.

123. One of the major products of fermentation that occurs in yeast is;
(1) phosphoglycolate.
(2) phosphoglycerate.
(3) ethanol.
(4) lactic acid.

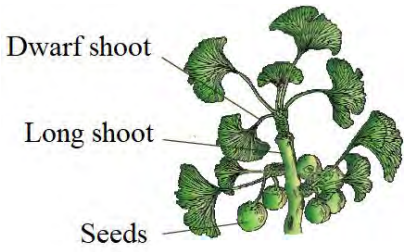
124. The plasma membrane of human RBC has;
(1) phospholipids as major lipids.
(2) approximately 52% lipids.
(3) approximately 40% proteins.
(4) equal amount of proteins and lipids.

125. Bacteria reproduce mainly by;
(1) sexual means.
(2) primitive type of DNA transfer from one bacterium to the other.
(3) fission.
(4) gametic fusion.

126. The cells of starch sheath in a dicot stem;
(1) are deposited with suberin.
(2) form the innermost layer of the cortex.
(3) give rise to lateral roots and vascular cambium during secondary growth.
(4) are barrel-shaped.

127. Identify the **true** statement about members of family Compositae.
(1) Placentation is usually axial.
(2) The fruit is caryopsis.
(3) Only one cotyledon is present in embryo.
(4) Flowers are bracteate.

128. Identify the **correct** statement about the plant shown below.



(1) It produces two types of spores.
(2) Its pollen grains are carried in water currents.
(3) The seeds are enclosed within fruits.
(4) The female gametophyte is unicellular.

129. Cells that do not divide further exit G_1 phase to enter an inactive stage called:
(1) G_2 phase. (2) M phase.
(3) G_0 phase. (4) S phase.

130. What is the net gain of ATP molecules for each molecule of glucose during fermentation?
(1) 38 (2) 36
(3) 8 (4) 2

131. Which plant growth regulator is used to speed up the malting process in brewing industry?
(1) Indole-3-acetic acid
(2) Gibberellic acid (GA_3)
(3) Abscisic acid
(4) Kinetin

132. The number of NADPH required for completion of one Calvin cycle is;
(1) 18 (2) 6
(3) 3 (4) 2

133. The protist that forms an aggregation called plasmodium which may grow and spread over several feet under suitable conditions is;
(1) slime mould. (2) a protozoan.
(3) a euglenoid. (4) a dinoflagellate.

134. The axoneme of eukaryotic cilia and flagella usually has _____ arrangement of microtubules.
(1) 9 + 0 (2) 9 + 2
(3) 9 + 1 (4) 4 + 4

135. Select the **correctly** matched pair representing the stem modification.
(1) *Opuntia* - Fleshy cylindrical photosynthetic stem
(2) *Euphorbia* - Flattened photosynthetic stem
(3) *Bougainvillea* - Thorns
(4) *Pistia* - Tendrils

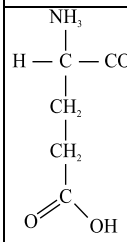
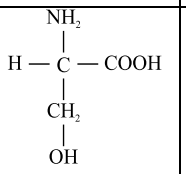
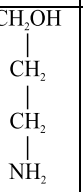
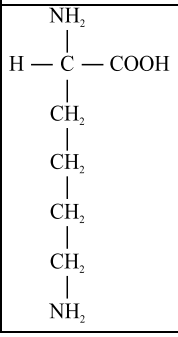
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136. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: All vertebrates are chordates, but all chordates are not vertebrates.
Reason R: The notochord is replaced by a cartilaginous or bony vertebral column in the adult stage of all chordates.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false and R is true.

137. Consider the following statements regarding secondary metabolites:
A. Some secondary metabolites like rubber and drugs have ecological importance.
B. Morphine and codeine are alkaloids.
C. Concanavalin A is an example of a toxin.
D. Anthocyanins are pigments that provide color to plant parts.
E. Cellulose and gums are polymeric substances.
How many of the above statements are **correct**?
(1) Two (2) Three
(3) Four (4) Five

138. Which of the following junctions help to stop substances from leaking across a tissue?
(1) Adhering junctions
(2) Gap junctions
(3) Tight junctions
(4) Intercalated discs

139. Which one out of A – D given below **correctly** represents the structural formula of the basic amino acid?

A	B	C	D
			

(1) B (2) C
(3) D (4) A

140. Match **List-I** with **List-II**.

List-I (Wave)		List-II (Electrical Activity)	
(A)	P-wave	(I)	Depolarisation of ventricles
(B)	QRS complex	(II)	Repolarisation of ventricles
(C)	T-wave	(III)	End of ventricular systole
(D)	End of T-wave	(IV)	Depolarisation of atria

Choose the **correct** answer from the options given below:
(1) A-IV, B-I, C-II, D-III
(2) A-IV, B-II, C-I, D-III
(3) A-I, B-IV, C-II, D-III
(4) A-II, B-I, C-IV, D-III

141. Nitrogenous waste is excreted in the form of pellet or paste by:
(1) *Pavo*
(2) *Ornithorhynchus*
(3) *Salamandra*
(4) *Hippocampus*

142. Regarding the mechanism of muscle contraction, choose the **correct** statements:
A. The distance between two 'Z' lines decreases during contraction.
B. The 'A' band retains its length during contraction.
C. The 'I' band increases in length during contraction.
D. The 'H' zone disappears during maximal contraction.
(1) A, B and C
(2) A, B and D
(3) B, C and D
(4) A and D only

143. Which part of the human brain contains the 'pneumotaxic centre' that can moderate the functions of the respiratory rhythm centre?
(1) Medulla oblongata
(2) Pons
(3) Cerebellum
(4) Thalamus

144. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Alpha-cells of Islets of Langerhans	(I)	Glucagon; Glycogenolysis
(B)	Beta-cells of Islets of Langerhans	(II)	Insulin; Glycogenesis
(C)	Zona glomerulosa	(III)	Aldosterone; Electrolyte balance
(D)	Zona fasciculata	(IV)	Cortisol; Anti-inflammatory

Choose the **correct** answer from the options given below:

(1) A-I, B-II, C-IV, D-III
(2) A-II, B-I, C-III, D-IV
(3) A-I, B-II, C-III, D-IV
(4) A-IV, B-III, C-II, D-I

145. Given below are two statements:

Statement I: The resting axonal membrane is nearly impermeable to sodium ions (Na^+) and negatively charged proteins in the axoplasm.

Statement II: The ionic gradient across the resting membrane is maintained by the active transport of 2 Na^+ outwards for 3 K^+ inwards.

In the light of above statements, choose the most appropriate answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

146. In which of the following phyla is the body divided into head, muscular foot, and visceral hump?

(1) Arthropoda
(2) Mollusca
(3) Echinodermata
(4) Annelida

147. What is the approximate percentage of the total cellular mass contributed by Nucleic acids?

(1) 70-90% (2) 10-15%
(3) 5-7% (4) 1%

148. Choose the **incorrectly** paired set:

(1) *Pheretima* - Metamerism
(2) *Asterias* - Water vascular system
(3) *Obelia* - Metagenesis
(4) *Adamsia* – Parapodia

149. In a person, the stroke volume is 50 ml and the heart rate is 100 beats per minute. What will be the Cardiac Output?

(1) 500 ml (2) 5000 ml
(3) 50 ml (4) 2000 ml

150. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Competitive inhibitors often resemble the substrate in their molecular structure.

Reason R: They bind to the active site of the enzyme and prevent the binding of the actual substrate.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false and R is true.

151. Which of the following structures is called the Sound Box in humans?

(1) Pharynx
(2) Larynx
(3) Trachea
(4) Epiglottis

152. The renal calculi are primarily composed of crystallised salts of:

(1) Sodium chloride
(2) Calcium oxalates
(3) Potassium carbonates
(4) Magnesium sulphates

153. Consider the following statements about the human skeletal system:

A. The cranium is composed of 8 bones.
B. The human skull is dicondylic.
C. There are 12 pairs of ribs, out of which the last 2 pairs are called vertebrochondral ribs.
D. The U-shaped hyoid bone is present at the base of the buccal cavity.
E. Each middle ear contains three tiny bones: Malleus, Incus, and Stapes.

Which of the statements are **correct**?

(1) A, B, C and E only
(2) A, B, D and E only
(3) B, C and D only
(4) A, B, C, D and E

154. Which hormone is responsible for the vigorus contraction of the uterus during child birth?
(1) Vasopressin (2) Oxytocin
(3) Progesterone (4) Relaxin

155. Nissl bodies are mainly composed of:
(1) Proteins and lipids
(2) DNA and RNA
(3) Nucleic acids and SER
(4) Free ribosomes and RER

156. Regarding the structure of proteins, identify the **correct** statements:
A. The sequence of amino acids is called the primary structure.
B. The first amino acid is the C-terminal amino acid.
C. In proteins, only left-handed helices are observed.
D. Tertiary structure is absolutely necessary for the biological activities of proteins.
(1) A and D (2) A and B
(3) B and C (4) A, C and D

157. Choose the **correct** match for the respiratory volume/capacity:
(1) Tidal Volume - 2500 ml
(2) Vital Capacity - TV + IRV + ERV
(3) Residual Volume - 500 ml
(4) Inspiratory Capacity - TV + ERV

158. Given below are two statements:
Statement I: In an open circulatory system, the blood is circulated through a closed network of blood vessels.
Statement II: Annelids and Chordates possess a closed circulatory system.
In the light of above statements, choose the most appropriate answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

159. Dialysing fluid used in hemodialysis has the same composition as plasma **except**:
(1) High glucose
(2) Nitrogenous wastes
(3) Essential minerals
(4) Proteins

160. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The mechanism of muscle contraction is explained by the sliding filament theory.
Reason R: Contraction takes place by the sliding of thick filaments over the thin filaments.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false and R is true.

161. Which of the following is a monosaccharide pentose sugar found in RNA?
(1) Glucose (2) Ribose
(3) Deoxyribose (4) Fructose

162. Given below are two statements:
Statement I: Juxtaglomerular apparatus (JGA) is formed by cellular modifications in the DCT and the afferent arteriole.
Statement II: A fall in GFR activates the JG cells to release Renin.
In the light of above statements, choose the most appropriate answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

163. In which part of the brain is the Corpus callosum located?
(1) Midbrain (2) Forebrain
(3) Hindbrain (4) Spinal Cord

164. Match **List-I** with **List-II**.

List-I (Hormone)		**List-II (Nature)**	
(A)	Insulin	(I)	Steroid
(B)	Epinephrine	(II)	Iodothyronine
(C)	Testosterone	(III)	Peptide/Protein
(D)	Thyroxine	(IV)	Emergency hormone and amino-acid derivative

Choose the **correct** answer from the options given below:
(1) A-I, B-II, C-III, D-IV
(2) A-III, B-IV, C-I, D-II
(3) A-III, B-I, C-IV, D-II
(4) A-IV, B-III, C-II, D-I

165. The "Hilum" is a notch found on the:
(1) Convex surface of the kidney
(2) Inner concave surface of the kidney
(3) Top of the adrenal gland
(4) Junction of the ureter and bladder

166. Which of the following is a characteristic of White Muscle Fibres?
(1) High myoglobin content
(2) Plenty of mitochondria
(3) High amount of sarcoplasmic reticulum
(4) Depend on aerobic process for energy

167. The enzyme Carbonic Anhydrase is found in:
(1) Plasma only
(2) RBCs only
(3) Both RBCs and Plasma
(4) Lymph only

168. Which of the following cells are the least abundant (0.5 - 1 percent) among WBCs?
(1) Neutrophils
(2) Basophils
(3) Monocytes
(4) Lymphocytes

169. The process of excreting ammonia is called Ammonotelism. Which of the following is **not** an ammonotelic animal?
(1) Bony fish (2) Aquatic insects
(3) Reptiles (4) Aquatic amphibians

170. Choose the **incorrect** pair regarding organism and Excretory organ:
(1) *Fasciola* - Flame cells
(2) Earthworm - Nephridia
(3) Cockroach - Malpighian tubules
(4) Prawn - Statocysts

171. Match **List-I** with **List-II**.

List-I (Phylum)		List-II (Feature)	
(A)	Ctenophora	(I)	Radula
(B)	Mollusca	(II)	Comb plates
(C)	Porifera	(III)	Water vascular system
(D)	Echinodermata	(IV)	Choanocytes

Choose the **correct** answer from the options given below:
(1) A-I, B-II, C-III, D-IV
(2) A-II, B-I, C-IV, D-III
(3) A-II, B-III, C-IV, D-I
(4) A-IV, B-I, C-II, D-III

172. Identify whether the following statements are **True** (T) or **False** (F):
I. All mammals (except a few) have seven cervical vertebrae.
II. Proteins are homopolymers formed from similar amino acids.
III. Heart failure is the same as cardiac arrest.
IV. Inulin is a polymer of fructose.
(1) T, F, F, T
(2) T, T, F, F
(3) F, F, T, T
(4) T, F, T, F

173. In which part of the nephron does the conditional reabsorption of Na^+ and water occur?
(1) PCT
(2) Descending limb of Henle
(3) DCT
(4) Ascending limb of Henle

174. What is the pH of human urine on average?
(1) 7.4
(2) 6.0
(3) 8.2
(4) 4.5

175. Match **List-I** with **List-II**.

List-I (Protein)		List-II (Function)	
(A)	Collagen	(I)	Hormone
(B)	Trypsin	(II)	Sensory reception
(C)	Insulin	(III)	Intercellular ground substance
(D)	Receptor	(IV)	Enzyme

Choose the **correct** answer from the options given below:
(1) A-III, B-IV, C-I, D-II
(2) A-IV, B-III, C-II, D-I
(3) A-I, B-II, C-III, D-IV
(4) A-II, B-I, C-IV, D-III

176. Which of the following is an example of an Acoelomate animal?
(1) *Ascaris* (2) *Taenia*
(3) *Nereis* (4) *Pila*

177. Myelin sheath is produced by:
(1) Schwann Cells and Oligodendrocytes
(2) Astrocytes and Schwann Cells
(3) Oligodendrocytes and Osteoclasts
(4) Osteoclasts and Astrocytes

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178. Given below are two statements regarding cockroach:
Statement I: The head holds a small proportion of a nervous system while the rest is situated along the ventral part of its body.
Statement II: A ring of gastric caeca is present at the junction of midgut and hind gut.
In the light of above statements, choose the most appropriate answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

179. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The human heart is myogenic.
Reason R: The normal activities of the heart are regulated intrinsically by specialised muscles (nodal tissue).
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false and R is true.

180. Frogs respire in water by skin and buccal cavity and on land by skin, buccal cavity and lungs.
Choose the **correct** answer from the following:
(1) The statement is true for water but false for land.
(2) The statement is true for both the environment.
(3) The statement is false for water but true for land.
(4) The statement is false for both the environment.

■■■

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